

JMod: Joint modeling of mass spectra for empowering multiplexed DIA proteomics

Kevin McDonnell,^{1,✉} Nathan Wamsley,¹ Jason Derks,¹ Sarah Sipe,¹ Maddy Yeh,¹ Harrison Specht,¹ & Nikolai Slavov^{1,2,✉}

¹Parallel Squared Technology Institute, Watertown, Massachusetts 02472, USA

²Department of Bioengineering, Northeastern University, Boston, MA 02115, USA

✉ Correspondence: kmcdonnell@parallelsq.org, nslavov@parallelsq.org

∈ Data, code & protocols: parallelsq.org/JMod

The throughput of mass spectrometry (MS) proteomics can be increased substantially by multiplexing that enables parallelization of data acquisition. Such parallelization in the mass domain (plexDIA) and the time domain (timePlex) increases the density of mass spectra and the overlap between ions originating from different precursors, potentially complicating their analysis. To enhance sequence identification and quantification from such spectra, we developed an open source software for Joint Modeling of mass spectra: JMod. It uses the intrinsic structure in the spectra and explicitly models overlapping peaks as linear superpositions of their components. This modeling enabled performing 9-plexDIA using 2 Da offset PSMtags by deconvolving the resulting overlapping isotopic envelopes in both MS1 and MS2 space. The results demonstrate 9-fold higher throughput with preserved quantitative accuracy and coverage depth. This support for smaller mass offsets increases multiplexing capacity and thus proteomic throughput for a given plexDIA tag, and we demonstrate this generalizability with diethyl labeling. By supporting enhanced decoding of DIA spectra multiplexed in the mass and time domains, JMod provides an open and flexible software that enables increasing the throughput of sensitive proteomics.

Introduction

Mass spectrometry (MS) empowers detailed analyses of protein abundances, higher-order structures, and post-translational modifications across proteomes^{1,2}. This versatile technique promises huge potential in the life sciences to investigate the functions of and interactions between different proteins³. The increasing sensitivity of MS has enabled both single-cell^{4–6} and single-organelle proteomics⁷. However, sensitivity and proteomic sequencing depth often compete against sample throughput: While analysis using isobaric mass tags can exceed 1,000 single cells per day^{8,9}, it has significant limitations, such as the ability to quantify only one peptide per mass spectrum. Currently, the throughput of alternative MS analysis rarely exceeds 200 samples per day — a severe practical limitation for many applications^{10,11}.

One prominent approach for increasing throughput is the parallelization of peptide fragmentation and sequence identification by label-free data-independent acquisition (DIA). The standard DIA method isolates and fragments the entire precursor m/z range over short segments in a repeating cycle^{12–15}. This technique parallelizes peptide acquisition by acquiring MS/MS spectra with wide quadrupole isolation windows to deliberately co-fragment multiple precursor ion species. This results in complex spectra and has motivated many algorithms for searching the data^{16–20}. DIA analysis algorithms often use one of two strategies to quantify precursors from convoluted mass spectra. First, software tools such as DIA-NN, EncyclopeDIA, and Spectronaut derive statistics from individual fragment and precursor ion chromatograms to identify precursors^{21–23}. These software tools use heuristics to find and remove interfered ion chromatograms from the quantitative analyses^{21,22}. Rather than discarding overlapping peaks, other DIA analysis tools, including Specter and CHIMERYs, model the mass spectra as a linear superposition of library fragment spectra^{24,25}. However these tools do not support parallelization by plexDIA or timePlex^{26,27}.

Proteomic throughput can be further increased by parallelizing samples using non-isobaric mass tags and multiplexed DIA (plexDIA)^{26,28,29}. Parallelizing along both dimensions increases throughput but also creates challenges for data analysis³⁰. Derks *et al.* used 3-plex non-isobaric mass tags to label each sample, providing a 3-fold increase in throughput²⁶. To further increase this throughput gain, plexDIA requires mass tags that can multiplex more samples.

Higher multiplexing is facilitated by tags offset by small mass differences³¹. Such tags have been used with data-dependent acquisition (DDA): Choi *et al.* demonstrated accurate MS1-based quantification with inter-plex spacing as close as 2 Da³². However, DDA does not parallelize the identification of peptides, sampling only one precursor at a time. Compared to DDA, DIA methods present at least two additional challenges for the analysis of mass-tagged samples with small offsets. First, a single wide quadrupole acquisition window in DIA captures several plexes of the same precursor species. The analysis must then identify each distinct plex from fragments with overlapping isotopic envelopes. Second, to realize the potential advantages of multiplexed DIA, the analysis software must quantify many lowly abundant precursors beyond the top N that may trigger MS/MS scans in a corresponding DDA method. The low-abundance signals are noisy and their isotopic envelopes can overlap, which together may inhibit correct peptide identification and compress the dynamic range.

A substantial challenge to multiplexing for any search algorithm is to maintain accurate identification and quantification of peptides and proteins with increasing data complexity. Even in label-free experiments, co-eluting precursor species may contribute precursor or fragment ions with identical or indistinguishable mass-to-charge ratios³³. This overlap increases with certain experimental parameters, such as shortening gradients or widening acquisition windows. Multiplexing further crowds the spectra in a specific pattern. If using mass tags with small mass offsets, the isotopic envelopes of adjacent plexes overlap with one another. As multiplexing increases, the proportion of non-overlapping peaks decreases, especially for mass tags with small mass offsets of less than 4 Da making methods that discard overlapping peaks suboptimal.

To meet these challenges and support multiplexing, here we introduce JMod, an open-source platform that models peak superpositions to enable increased DIA multiplexing. The model is designed to deconvolve the structured overlap that arises in both MS1 and MS2 spectra from DIA acquisition of samples multiplexed by mass tags (plexDIA), by sample-specific retention time offsets (timePlex), or by both combined. JMod uses a linear superposition model to leverage structure in the data created by multiplexing to improve sequence identification and to resolve quantitative differences between overlapping precursor and fragment ions. Moreover, as an open-source software tool, JMod can serve as a community platform for developing new computational

workflows and capabilities.

Results

Frequency of overlapping fragments

In DIA approaches, many precursors are isolated and fragmented together and the resulting fragments may have mass-to-charge ratios that are identical or too close to be resolved by the instrument. This introduces ambiguity in precursor-to-fragment associations and thus challenges identification and quantitation. To investigate the extent of this overlap in typical proteomic use cases, we calculated the theoretical fraction of fragments from each entry in an empirical spectral library of 209,712 precursors derived from analyzing human samples that would overlap with the fragments from the other entries (Fig. 1a). To calculate this, we used a m/z tolerance of 0.01 Th and a retention time tolerance equal to 1.5% of the total library range.

DIA approaches utilize a variety of window widths for collecting more or fewer precursors in parallel, with their utility dependent on sample input amount or complexity. For a label free experiment, precursors were found to have on average fewer than 70% distinct fragments when using 16 Th isolation windows. This fraction decreases as the window size increases. Similarly, parallelizing data acquisition to increase throughput increases the potential for fragment overlap. Our estimates show that parallelizing in the mass domain using plexDIA further decreases the fraction of distinct fragments, even for acquisition strategies using 16 Th wide MS2 windows. For a 9-plex experiment, most precursors have less than 15% of their fragments being distinct, with many having no unique fragments at all. This highlights the need for approaches that model peak overlap if multiplexing capabilities are to be increased.

Joint modeling of MS2 spectra

Fragment overlap can be reduced by further reducing the isolation window width, although this also reduces the fraction of captured ions and thus sensitivity^{34,35}. An alternative solution is to model this overlap and therefore account for the multiple sources which can contribute to an ob-

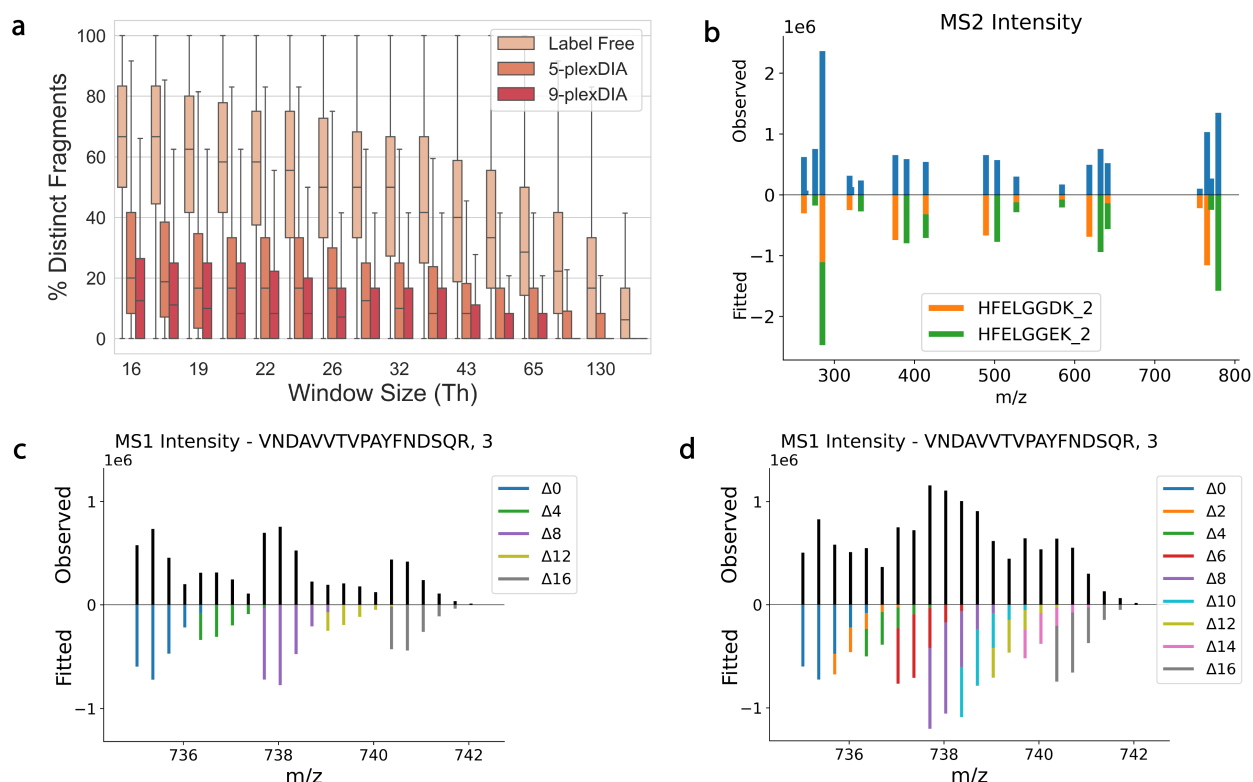


Figure 1 | Modeling overlapping peaks in DIA mass spectra. **a** Distributions of the percentage of each precursor's monoisotopic fragments that are distinct (not overlapping with other monoisotopic fragments) for label free, 5-plexDIA, and 9-plexDIA. All fragments were derived from a library of 209,712 tryptic human precursors and compared using a m/z tolerance of 0.01 Th and a retention time tolerance of 1.5% of the total range. y1 ions were not considered. **b** Mirror plot of the observed and fitted fragment intensities for 2 co-eluting precursors; (HFELGGDK, 2+) and (HFELGGEK, 2+). While JMod also models the isotopic distributions of each fragment, only peaks that match the monoisotopic fragments of the precursors are shown to enhance clarity. **c** Mirror plot of the observed (top) and fitted (bottom) MS1 intensities for charge 3+ VMDAVVTVPAYFNDQR precursors labeled with 5-plex PSMtags³¹ spaced by 4 Da. **d** Mirror plot similar to panel c but with 9-plex PSMtags spaced by 2 Da.

served ion.

Towards this goal we developed JMod, an open-source software tool that supports increasing multiplexing in mass spectrometry proteomics through the joint modeling of precursors and their fragmentation spectra. JMod models the empirical MS1 and MS2 mass spectra as linear combinations of all peaks (inducing isotopic envelopes) in the spectral library that can contribute to the measured intensities. It can therefore identify precursors that share many of the same fragments as shown in Fig. 1b.

JMod uses quadratic programming to find the set of non-negative coefficients that minimize

the squared distance between the superposition of the library fragments and the observed peaks. Despite sharing many of their fragments, both precursors in Fig. 1b can be identified as their superposition minimizes the residual error to the observed spectrum. Furthermore, the joint modeling approach taken by JMod finds the minimum set of precursors that can best explain the spectra, so precursors that spuriously match fragments will not be given a significant coefficient by the model.

Increasing multiplexing by modeling overlapping isotopic envelopes

While analyzing mass spectra with overlapping peaks presents challenges, it can confer substantial benefits if the overlap is successfully modeled. Specifically, it enables increasing the capacity for sample multiplexing. To evaluate this potential, we used JMod to model intentionally engineered overlap between the isotopic envelopes of samples labeled with mass tags that have channel offsets of both 4 and 2 Da (Fig. 1c and Fig. 1d). Channels with smaller offsets result in highly overlapping isotopic envelopes that can be modeled naturally using JMod. To do this, JMod calculates the sequence-specific isotope distributions for each labeled precursor using its amino acid composition, while also accounting for the peculiar isotopic distribution of each mass tag. Then, JMod finds the set of MS1 coefficients that minimize the residuals between the observed MS1 peaks and the theoretical isotopic envelopes. The same approach is applied to modeling the fragments and the overlap between their isotopic envelopes.

Modeling peak superpositions enables increased sample throughput as it can deconvolve the relative contributions of mass tag channels that have overlapping isotopic distributions. Without modeling overlaps, methods are limited to using tags separated by at least 4 Da to reduce the overlap between isotopic envelopes^{26,30}. However, even for channels separated by 4 Da, contributions from isotopes in lighter channels can be significant, especially if the peptides labeled with the lighter tags are more abundant, as seen in Fig. 1c. By modeling these channels simultaneously, JMod aims to quantify each sample independent of the overlap. This allows the reduction of the spacing between channels while maintaining the ability to deconvolve the signal (Fig. 1d).

This isotopic overlap also occurs at the MS2 level. Except possibly for fragments containing internal lysine residues, each tagged fragment will be spaced from the adjacent mass channels by the mass shift of a single tag. Furthermore, the y-ions of arginine-terminating precursors will be

indistinguishable across plexes for tags such as PSMtag³¹, which use amine labeling chemistries. JMod can account for this MS2 isotopic overlap by predicting fragment specific isotope distributions for every precursor and including these in the model.

Other methods to increase throughput will further increase the density of spectra and therefore peak overlap. Multiplexing in the mass domain (‘plexDIA’) can be complemented by multiplexing in the time domain (‘timePlex’)²⁷. Derks *et al.* demonstrated this by acquiring multiple samples with staggered chromatographic separations where these time differentials encode sample specificity. The orthogonality of time and mass domains enables multiplicative scaling when used together. This new multiplexing technique is uniquely supported by JMod.

Using these modeling approaches, JMod supports four modes of data acquisition: label-free DIA (no sample multiplexing), plexDIA²⁶ (multiplexing with mass offsets), timePlex²⁷ (multiplexing in time), and combined plexDIA and timePlex. For plexDIA acquisition, JMod provides native support for mTRAQ, diethylation, and PSMtag³¹. In addition, users can create their own tags and add them to the tag library.

Benchmarking sequence identification

We first evaluated sequence identification by JMod using entrapment on label-free samples containing mixed human and yeast proteomes. To model single-cell level input, human and *Saccharomyces cerevisiae* tryptic protein digests were diluted to 200 pg and 80 pg respectively per injection. We compare JMod to DIA-NN²¹, which is the leading alternative for processing both label free and plexDIA data that is also freely available to the research community. To compare precursor and protein-group identification rates at the same stringency threshold, we plotted the number of identified human precursors and proteins by each method as a function of the empirically derived entrapment false discovery rates (eFDR), Fig. 2a. While the inclusion of entrapment sequences in the library reduces the total number of identifications for both algorithms, their relative performance can be evaluated at a 1% eFDR²¹. The eFDR is calculated using the “paired method” as described by Wen *et al.*³⁶, where every target sequence in the library has a corresponding shuffled entrapment sequence. In this analysis, JMod achieved a comparable number of identifications, particularly at high confidence.

We then sought to evaluate the false discovery rates assigned by JMod to identified precursors by comparing them to the eFDR, Fig. 2b. The FDR and eFDR are compared for 3 technical replicates of the label-free data. For these label-free data, JMod reports a controlled estimate of the FDR, more conservative than those derived empirically.

Next we sought to evaluate the FDR control of JMod for highly multiplexed samples. Toward increased multiplexing, JMod enables, for the first time, searching DIA data from samples with substantially overlapping isotopic envelopes. As test samples, we used 9-plex sets labeled with PSMtags³¹ which have channels separated by just 2 Da. The sets contained mixed species samples described in Table 2. Each sample contained 200 pg human digest as well as *Saccharomyces cerevisiae* digest ranging from 20 to 80 pg. We evaluated the search results using an entrapment analysis with shuffled sequences. The results in Fig. 2c show that the empirically derived precursor FDR estimated by JMod is an accurate representation of the eFDR for the 3 technical replicates of the 9-plex data. This indicates that the reported false discovery rates were well-calibrated for multiplexed data with overlapping isotopic envelopes.

Throughput is proportional to plex-size

As the number of samples analyzed in parallel increases, the number of quantified proteins per sample should remain the same, thus increasing the number of quantitative data points proportionally to the plex. We evaluated the ability of JMod to support such scaling for increasing the plex-size using PSMtags³¹. The results in Fig. 2d show that both JMod and DIA-NN support 5-fold scaling with the 5-plex PSMtags offset by 4 Da. For both algorithms, the coverage per sample is not strongly affected by multiplexing and the number of proteins identified per run scaled with the plex-size.

We further evaluated JMod's ability to identify and quantify PSMtag labeled peptides in a 9-plex configuration where adjacent mass channels differed from one another by 2 Da offsets (Fig. 2d). Currently, 2 Da channel offset DIA data are only searchable using JMod. This reduction in channel spacing provides almost twice the potential throughput for PSMtags increasing the plex-size from 5 to 9 without any changes to the molecule. We used Jmod to search these data and then compared the results to both the $\Delta 0$ and 5-plex configurations to evaluate performance.

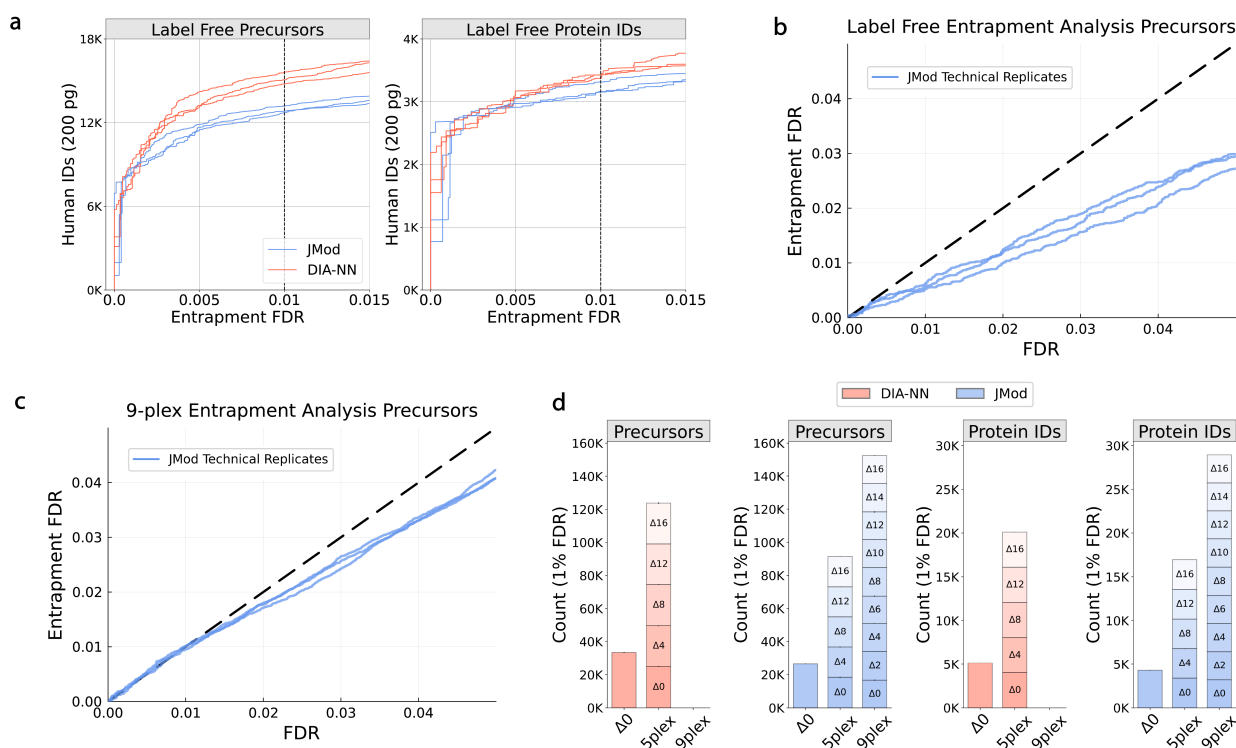


Figure 2 | Evaluating sequence identification from label free and multiplexed DIA data. **a** The identification performance of JMod (blue) and DIA-NN (red) on a 200 pg label-free sample. The number of human precursors and proteins is plotted versus the entrapment FDR for 3 replicate injections. **b** The q-values reported by JMod versus the empirically derived q-values for precursors identified in the 200 pg label-free entrapment experiment. **c** The q-values reported by JMod versus the empirically derived q-values for precursors identified in the 200 pg per channel, PSMtag 9-plex entrapment experiment. **d** Human identification rates for JMod (blue) and DIA-NN (red) from single and multiplexed PSMtag experiments. Both JMod and DIA-NN were used to search the $\Delta 0$ and 5-plex data. Only JMod supports searching of the 9-plex data with 2-Da channel offsets.

For the 9-plex experiment with 2 Da separation, the number of proteins per sample again scaled with the plex-size, providing similar coverage per sample as non-multiplexed and 5-plex multiplexed samples. This resulted in almost 30,000 protein data points from 9 single-cell level samples analyzed in a single run.

Quantifying overlapping isotopic envelopes

By modeling the superposition of isotopic envelopes, JMod is able to quantify non-isobarically multiplexed precursors separated by less than 4 Da. To achieve this, JMod starts by predicting the isotopic envelope for each ion from its sequence and then finds the superposition of isotopic

envelopes with associated coefficients that best explain the measured intensities. This provides a coefficient for each precursor in each channel for each spectrum. These can be combined to estimate the elution profile of each peptide over time. This is exemplified in Fig. 3a for the yeast peptide TNNPETLVALR labeled with PSMtags in both the 5-plex and 9-plex schemes. JMod is able to produce consistent elutions for the same channels and inputs despite there being much more overlap between isotopes in the 9-plex scheme. Jmod then calculates the area under these coefficients over time to quantify the peptide in each channel at the MS1 level.

Repeatability of highly multiplexed protein measurements

To assess the repeatability of protein quantitation for the 9-plex channels, we evaluated the coefficient of variation (CV) of the proteins confidently identified in each channel with at least 2 precursors. We also compared these to the CVs from similar label-free data searched using JMod, Fig. 3b. Each channel in the 9-plex data had a median protein CV in the range 0.07-0.11 for the intersected proteins. The equivalent label free proteins had a median CV of 0.09. Both the label free and 9-plex values are well within the range of acceptable CVs, indicating repeatable quantitation for these highly multiplexed samples.

Accurate quantitation for overlapping isotopic envelopes

We next evaluated the accuracy of quantitation performed by JMod against mixing ratios of human and yeast proteomes at approximate single-cell-level inputs (Table 2). Mixtures labeled with 5-plex PSMtag were searched by both JMod and DIA-NN while 9-plex mixed proteomes were uniquely searched using JMod. To fairly assess the differences between the separate models and offsets, only precursors identified in all 3 analyses were used for the comparison. We specifically focused on ratios involving mixture A (Table 2), which has the smallest input amount of yeast. This means it has the largest fold changes relative to other mixtures, but also its precursors are more lowly abundant making them harder to quantify. The relative ratios of the precursors identified in the different conditions were then compared to the expected mixing ratios for each channel. JMod was found to perform best for the 5-plex configuration where all channels are separated by at least

4 Da (Fig. 3c). However, both JMod and DIA-NN exhibited high accuracy of quantitation, with distributions tightly centered around the expected mixing ratios.

We then compared the quantitative accuracy of JMod for the 9-plex (separated by 2 Da) for the same intersection of precursors. In this comparison, JMod showed similar high accuracy for the 5-plex and 9-plex samples despite the increased peak overlap and additional sample complexity with the 9-plex. Although the distributions of ratios exhibited a slightly wider spread than the 4 Da spaced data, each was centered on the expected ratio, indicating high quantitative accuracy.

We also performed a systematic comparison of the MS1 quantitative accuracy between the different setups. Fig. 3d shows the distribution of absolute precursor ratio errors for those shared between DIA-NN and JMod for the different channel offsets and plex sizes. JMod and the 4 Da spaced 5-plex data was found to be the most accurate setup, with the smallest median error of the three. However, all three setups showed high accuracy for the vast majority of the shared precursors.

To more quantitatively compare the impact of the reduced mass offset, we focused on precursors that have a single tag and are thus separated by smaller offsets. We therefore systematically compared the quantitative performance of JMod for arginine-terminating precursors in the 5-plex, 4-Da spaced data and the 9-plex, 2-Da spaced data, Fig. 3e. In this case, the absolute error between the observed $\log_2$ ratio and the mixing ratio was calculated for all precursors from both human and yeast. These errors were then compared between the 4-Da and 2 Da offset data, separated by MS2 and MS1 quantitation (Fig. 3e). JMod exhibited high and comparable accuracy between the 4-Da (dark gray) and 2-Da (light gray) schemes across all conditions, with very little difference in median absolute errors.

MS1 accuracy tends to be higher for these low input amounts⁷, as was the case here. At the MS2 level, the y-ions of all arginine-terminating precursors in all channels are shared and the b-ions are separated only by 2 Da. As such, these precursors were the most difficult to quantify and exhibited the largest absolute errors. Nonetheless, the errors for the 5-plex and 9-plex samples are similar, supporting the ability of JMod to extend plexDIA quantification to tags separated by smaller mass offsets.

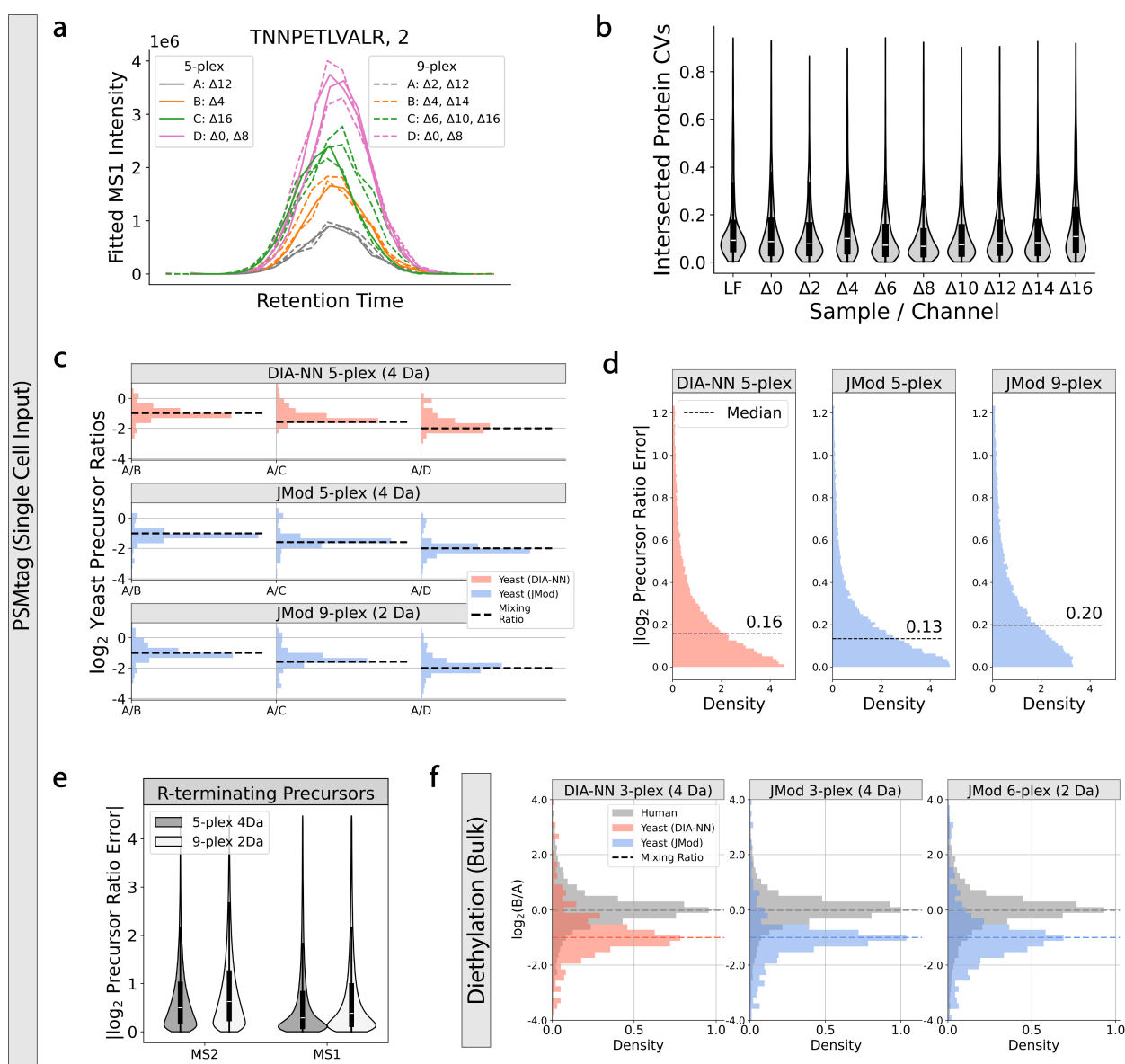


Figure 3 | Evaluating quantitative accuracy for plexDIA using 4-Da and 2-Da offsets. **a** Fitted MS1 coefficients for the same yeast precursor (TNNPETLVALR, 2+), in all channels in a 5-plex and 9-plex acquisition. Colors indicate the different mixtures and therefore input amount. **b** Comparison of repeatability of quantification with JMod between the intersected proteins from the label free experiment and all channels in the 9-plex experiment. **c** Distributions of the observed ratios versus the mixing ratios for all yeast precursors identified by DIA-NN and JMod when searching the 5-plex PSMtag data as well as identified by JMod in the 9-plex data. To enhance the clarity of the plot, human ratios were excluded, and channel ratios were grouped by their mixture label (A-D). **d** Arginine precursors are separated by 4 Da in the 5-plex scheme and by 2 Da in the 9-plex scheme. The absolute ratio error for all arginine-terminating precursors is compared between the 5-plex and 9-plex schemes at both the MS1 and MS2 level. **e** The MS1 ratios for diethyl labeled human and yeast precursors identified by both DIA-NN and JMod when searching the 3-plex data separated by 4 Da, as well as identified by JMod when searching the 6-plex data separated by 2 Da.

Generalizability of JMod across instruments and mass tags

JMod can be used with different tags and MS instruments. To demonstrate this, we evaluated the quantitative accuracy of JMod when searching diethylated bulk samples separated by both 4 Da and 2 Da. Diethylation can provide multiplexing capabilities of 6-plex if allowing just 2 Da offsets between channels. However, deuteration of the diethyl tags introduces retention time shifts between mass channels that complicates quantitation³⁷.

We again performed a mixed species experiment with mixed human and yeast proteomes this time using an Exploris 480 instrument. Bulk mixed human and yeast proteomes were labeled using diethylation with both a 4-Da spaced 3-plex scheme and a 2-Da spaced 6-plex scheme. Fig. 3e shows the precursor ratios for those identified by both DIA-NN and JMod in the 3-plex diethylation scheme and also by JMod in the 6-plex diethylation scheme. The comparison between the measured and the mixing ratios demonstrates comparable and accurate quantification for JMod for both 4-Da and 2-Da spaced schemes. JMod does not require channels to elute at the same time and can model the isotopic overlap across the elution of all channels despite the addition of retention time effects. The partial overlap can therefore be modeled with the area under the MS1 coefficients still providing accurate quantitation.

Discussion

Biomedical research needs scalable and affordable analysis of many thousands of samples across diverse applications ranging from population-scale plasma proteomics to single-cell proteomes^{8,38,39}. A promising avenue for achieving such scalability is to parallelize data acquisition⁴⁰. However, such approaches increase data density and lead to increased frequency of overlapping MS1 and MS2 signals, Fig. 1a. We developed JMod, an open source platform, to model these challenges and support scaling of MS proteomics by parallelization.

JMod seeks to support all forms of parallelization by modeling the predictable structure of multiplexed DIA data. Here, we demonstrate unique support for increased overlap between channels of non-isobaric mass tags. This allows the use of smaller mass offsets and therefore enables about 2-fold higher throughput with PSMtags. Similarly, JMod supports sample parallelization in

time by introducing small delays between sample separations and acquiring data from the overlapping active gradients by DIA. This method, termed timePlex²⁷, is directly and uniquely enabled by JMod.

Beyond supporting highly parallelized data acquisition, JMod offers advantages for the analysis of peptides with similar amino acid sequences as it can model their overlapping fragments as shown in Fig. 1b. This advantage should benefit peptides from post-translationally modified proteins and proteoforms originating from alternate RNA decoding⁴¹. JMod can also support increasing levels of multiplexing by metabolically incorporated heavy amino acids offset by smaller mass differences, which can increase the throughput of quantifying protein synthesis and degradation rates^{42,43}.

The joint modeling approach of JMod provides highly accurate quantitation of precursors from single-cell-level input amounts regardless of the channel overlap (Fig. 3c). For 4 Da spaced channels, which are supported by other platforms, JMod performed better than the state-of-the-art software DIA-NN (Fig. 3d). While for the same data, DIA-NN currently identifies more precursors, we anticipate improved sensitivity in future versions of JMod.

This accurate and precise modeling of fragment intensities requires spectral libraries that accurately reflect fragmentation patterns in the data. This can be a limitation if the fragmentation patterns are challenging to measure or predict, as in the case of newly developed tags, such as PSMtags. However, this limitation is greatly mitigated by the many high quality empirical libraries and predicated libraries from deep learning models^{44–46} and refined libraries from transfer learning¹⁸. We expect to increase the accuracy and sensitivity of JMod for the data described in this work when better models have been generated.

The use of tags with 2-Da offsets offers advantages and limitations. For a given molecular composition of the tags, JMod's support for reduced mass offsets increases the maximal plex that can be designed. Furthermore, the overlapping peaks decrease the total number of peaks for a given plex, which reduces the potential for interference by close peaks compared to the same plex achieved with 4-Da mass tags. A downside of 2-Da spaced tags is that measurement errors can limit the ability to quantify very large fold changes, such as 100-fold. We did not encounter such limitations within the tested range showing in Fig. 3c.

We expect to continue building new features into JMod for enhancing sequence identification and quantification from highly parallelized DIA data. These features will support new experimental designs, as in the case of timePlex, support more data types (including ion mobility), and further exploit the inherent and engineered structure of multiplexed data⁴⁷. We hope that the open source nature of JMod will encourage others to contribute to these future developments.

Data and Code availability: The source code for JMod is available at:

github.com/ParallelSquared/JMod. Raw data, spectral libraries, search outputs, and meta data are available at MassIVE: [MSV000097957](https://massive.ucsf.edu/MSV000097957). The code used to generate the figures is available at: github.com/ParallelSquared/JModPreprint

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Competing Interests: J.D., K.M.D., H.S. and N.S. are listed as inventors on a patent application for timePlex. H.S., S.S., M.Y. and N.S. are listed as inventors on a patent application for PSMtag.

Author Contributions

Experimental design: K.M.D., H.S. and N.S. **Sample preparation and acquisition:** S.S. and M.Y. **Raising funding & supervision:** H.S. and N.S. **Data analysis:** K.M.D., N.W. and J.D. **Writing & editing:** K.M.D., N.W., S.S. and N.S.

Methods

Sample Preparation

The two-proteome samples were prepared from human K562 proteins (Promega, V6941) and *S. cerevisiae* (yeast) peptides (Promega, V7461). Intact proteins were provided in a buffered urea solution to which dithiothreitol then iodoacetamide were added to reduce and alkylate cysteine residues. The urea was diluted with 120 mM ammonium bicarbonate (ABC) then digested with trypsin overnight at 37°C using a 1:50 enzyme:protein ratio. Resulting peptides were desalted using C18 cartridges on a negative pressure manifold. Human and yeast peptides were added at quantitative ratios as demonstrated in Table 1 for diethyl labeling. Alternatively, predigested human peptides (Promega, V6951) were combined with yeast peptides at ratios shown in Table 2 for PSMtag labeling.

Diethylation

The diethylation protocol was adapted from previous reports^{32,48,49}. Dried peptides were reconstituted in 100 mM HEPES buffer to 500 ng/ μ L. Light (C_2H_4O), medium ($^{13}C_2H_4O$), and heavy (C_2D_4O) acetaldehyde was added from a 20% v/v stock to final concentrations of 2.5%. Sodium cyanoborohydride ($NaBH_3CN$) and sodium cyanoborodeuteride ($NaBD_3CN$) were prepared fresh daily and added to final concentrations of 250 mM reducing agent. Isotopes of acetaldehyde and the reducing agents were combined to obtain six possible masses. Two additional masses can be achieved with a ‘heaviest’ form of acetaldehyde, $^{13}C_2D_4O$, in combination with the two reducing agents, which would allow 4-Da spaced 4-plex or 2-Da spaced 8-plex.

The diethylation reaction proceeded for 1 hr at 37°C with shaking at 1000 rpm. Labeling was quenched for 10 min with ABC at a final concentration of 500 mM. The final peptide solution was acidified with formic acid, cleaned using house-packed C18 spin tips, and then dried. The dry peptides were reconstituted with mobile phase A (MPA, 0.1% formic acid in water), combined, and subjected to LC-MS/MS analysis in either a 3-plex or 6-plex combination.

Table 1 | Mixing ratios for diethyl labeled two-proteome experiment.

Channel	$\Delta 0$	$\Delta 2$	$\Delta 4$	$\Delta 6$	$\Delta 8$	$\Delta 10$
Condition	A	C	B	D	A	C
Human (ng)	100	100	100	100	100	100
Yeast (ng)	6	18	12	24	6	18

PSMtag Labeling

PSMtag employs NHS-ester chemistry to label primary amines present on peptide N-termini and lysine residues. Samples were each labeled with one of nine non-isobaric isotopologues of PSMtag: the lightest form, $\Delta 00$, and eight heavier forms increasing in mass by 2 Da each up to $\Delta 16$. All PSMtag analogues were synthesized and provided in powder form by CreaGen, Inc. Tags were solubilized in anhydrous DMSO to a concentration of 24 mg/mL. An organic base, N,N-diisopropylethylamine (DIPEA), was combined with the tag stocks to final concentrations of 4% (approx. 218 mM) DIPEA and 12 mg/mL PSMtag in DMSO. This labeling solution was added to dried peptides at a ratio of 1.5:1 tag to peptide mass. The reaction was allowed to proceed at room temperature for 2 hr prior to quenching with hydroxylamine to a concentration of 0.25%. Samples were directly diluted with 30% acetonitrile, 0.18% n-dodecyl- β -D-maltoside (DDM), and 2% formic acid prior to LC-MS/MS analysis.

Samples used for benchmarking

We combined 4 different mixtures of proteome digests from *Homo sapiens* and *Saccharomyces cerevisiae* at precise ratios (Table 2). Human proteomes were kept at a constant input for each channel (200 pg). Yeast proteomes were added at varying mixture amounts, providing abundance ratios of 1:1 up to 4:1 between the different samples. These mixtures were then labeled with different channels of the newly developed non-isobaric tag, PSMtag³¹.

Table 2 | Mixing ratios for PSMtag two-proteome experiment

Channel	$\Delta 0$	$\Delta 2$	$\Delta 4$	$\Delta 6$	$\Delta 8$	$\Delta 10$	$\Delta 12$	$\Delta 14$	$\Delta 16$
Condition	D	A	B	C	D	C	A	B	C
Human (pg)	200	200	200	200	200	200	200	200	200
Yeast (pg)	80	20	40	60	80	60	20	40	60

LC-MS Acquisition

Bulk DE-proteome analysis

For demonstration of quantitative accuracy and precision of JMod for plexDIA data, diethylated peptide samples were injected as a 4-Da spaced 3-plex or a 2-Da spaced 6-plex. At least 100 ng per channel were injected on an IonOpticks Aurora Ultimate C18 column using a Vanquish Neo LC coupled to an Exploris 480 Orbitrap MS (Thermo Scientific). Peptides were loaded using 5% mobile phase B (MPB, 80% acetonitrile with 0.1% formic acid) and separated with a gradient from 10% MPB to 36% over one hour using a flow rate of 200 nL/min. MPB increased to 95% and remained for 8 min before column reequilibration back to 6% MPB. Orbitrap resolutions of 120,000 and 30,000 were used for MS1 and MS2 scans, respectively. A variable window scheme with 60 MS2 scans ranging from m/z 380 to 1370 all overlapping by 0.5 Th; the first 38 windows were 8.5 Th wide, the following 9 were 17.5 Th, and the final 13 were 41.5 Th wide. A constant normalized collision energy of 30% was applied with maximum injection time set to “Auto” to be synchronized with Orbitrap acquisition duration.

PSMtag data acquisition

PSMtag-labeled samples were analyzed with a Vanquish Neo LC and an Orbitrap Astral MS equipped with a FAIMS Pro Duo source. Samples were loaded using 30% MPB. Separation occurred from 38% MPB to 62% for 24.5 min and to 75% MPB for another 5.4 min using a flow rate of 100 nL/min. Peptides were ionized with an electrospray ionization voltage of 1900 V and introduced to the FAIMS source, which implemented a constant -50 V compensation voltage and 3.6 L/min gas flow rate. MS1 spectra were collected with an AGC of 300%, maximum injection

time of 150 ms, and resolving power of 240,000. MS2 spectra were collected using 20 variable windows from m/z 500 to 1,000 in the Astral with an AGC of 500%, injection time of 60 ms, and normalized collision energy of 24%.

Library generation

A diethylated peptide library containing human and yeast proteome sequences was generated from a gas-phase fractionated, data-dependent acquisition experiment using a Vanquish Neo and Orbitrap Astral. Two fractions from m/z 300-600 and from m/z 600-1280 were collected in duplicate from 500-ng of peptide input each. MS/MS precursors were filtered by charge states 2-5. Data were searched in SpectroMine (v4.5) and the library was exported to include up to 40 fragments per precursor. The final library included over 69k human and 11k yeast precursors.

The label free library used in the experiments was generated from searching a bulk dataset using DIA-NN 1.8.1. Three replicates containing 250 ng each of human Promega K562 and Promega yeast tryptic peptide digests were acquired in DIA mode on an Exploris 240. These mixtures were searched with a DIA-NN generated predicted library of both the human and yeast proteomes, with the resulting library being used. Carbamidomethylation of cysteine was considered as a fixed modification. The entries in this library were then appended with their shuffled counterparts as described in the entrapment section below.

PSMtag libraries were created using AlphaPeptDeep⁵⁰. Training data were made up of precursors identified in bulk Human digests acquired in DDA mode on an Orbitrap Astral and searched using Comet. Following training, the resulting model was then used to predict the full human and yeast proteomes. Mixed Promega K562 and Promega yeast digests labeled with PSMtag data were then acquired in DIA mode on the Orbitrap Astral. The AlphaPeptDeep predicted full proteome library was used to search this PSMtag data with DIA-NN 2.1 to create a smaller PSMtag library for use in the low-input experiments.

The JMod Algorithm

Data input and processing

JMod accepts centroided data in mzML format and libraries in a tab delimited format. If specified, JMod will add additional fragment isotopes to the library using the BrainPy package⁵¹.

For best results, JMod utilizes Biosaur2⁵² feature identification software to assist with the first search. Biosaur2 clusters peaks from MS1 spectra into those originating from a single peptide's isotopic envelope. This provides a list of features with specific m/z and retention times from potential highly abundant peptides. These features are only used for alignment and are not used in the main search but help provide preliminary high confidence IDs. If features are not supplied, JMod instead searches the n spectra with the largest total ion current across the gradient.

Preliminary search

JMod uses a preliminary search to align the library to the observed spectra. This helps account for differences between library retention times and observed retention times as well as errors in measured peak m/z . During the preliminary search, the Biosaur2-identified peptide-like features are used to discern potential highly abundant precursors across the m/z range and gradient. MS2 spectra closest to the apex of each feature are searched for possible library candidates matching the precursor m/z using a wide (20 ppm) tolerance. These identifications are then sorted by how well they fit based on attributes collected during the preliminary search. Starting with precursors where all their attributes rank above the 20th percentile, JMod iteratively increases this threshold until 95% of the identifications' retention time errors fit a gaussian error distribution about the line of best fit. This step removes low-scoring false positives, preventing them from impacting the alignment.

Retention time alignment and fine-tuning

JMod also employs machine learning to fine-tune the library to the experimental conditions using these precursors. This is similar to an approach introduced by Wallmann *et al.*¹⁸. High-scoring

peptides are split into training and validation sets. JMod then uses these high-scoring potential peptides and their observed retention times to fine-tune its internal, pretrained, convolutional neural network (CNN) models. This step includes early-stopping to prevent the models from overfitting. Fine-tuning provides much better predictions of observed retention times, allowing JMod to use a much more strict retention time tolerance. Depending on the experiment, JMod automatically selects one of four different base models: these are label-free, mTRAQ, diethylation, and PSMtag. A user can also fine-tune a library on their own data and use this for their subsequent searches instead of the base models. This is particularly useful for low-input samples where there may not be enough preliminary IDs to perform adequate fine-tuning.

Once fine-tuning is complete, the updated predictions for the validation set and the original library retention times are compared to observed values. JMod will select the fine-tuned predictions, only if they better describe the data. Otherwise, the original retention times will be used. Users can also skip the fine-tuning step if they know that the library contains accurate empirical retention times.

The area under the cumulative distribution of the absolute retention time errors is used to decide on the best model. Those with smaller errors will have a larger area under this curve. The error distributions are modeled as both Gaussian and exponential distributions, with the best approximation being used to derive optimized retention time tolerances.

The identified precursors are also used to correct for any systematic m/z errors. A LOWESS regression is first used to correct for changes in m/z error across retention time. A subsequent LOWESS regression line is used to account for m/z errors that depend on the m/z itself. An m/z tolerance is then calculated as 4 standard deviations of the Gaussian distribution fit to the MS1 errors from the preliminary search.

Main search

Before the main search, the library retention times and precursor m/z are adjusted to match the experiment using the models derived in the first search. If performing plexDIA or timePlex experiments, the library is also duplicated for each channel, with each receiving unique retention times and m/z values. JMod natively supports multiple labeling schemes including PSMtag, mTRAQ,

and diethylation, while users can also add custom tags to the tag library. Decoys are then generated for each precursor in the library using a pseudo-reverse method⁵³. The data are then searched using the combined target and decoy libraries.

The main search models each MS2 spectrum independently as a linear superposition of the library precursors. Library precursors that fall within the retention time tolerance and window m/z range are considered as potential candidates. Those candidates that do not match their precursor m/z to a peak in the nearest MS1 scan are then excluded, along with those that match less than 50% of their fragment intensity. These parameters can be adapted to the experiment in question if required. The remaining precursors are used to set up non-negative least squares problem in the form:

$$\min_{\mathbf{x} \geq 0} \|\mathbf{Ax} - \mathbf{b}\|_2^2 \quad (1)$$

Each column of the matrix A represents a potential precursor to be considered as part of the observed spectrum. The observed peak intensities are represented by b . Each row of A then represents the library fragments that match the m/z of the corresponding row in b . If a precursor does not have a fragment that matches a particular peak, the entry in that row is a zero, while if it is a potential match, the entry is the normalized fragment intensity. To account for library fragments that do not match any peak in the spectrum, zero entries are added to the end of b ; one for each fragment that has no corresponding match. This penalizes library spectra that have many fragments that are not matched to the spectrum. The values of x that minimize Equation (1) are obtained using the quadratic programming solver from CVXOPT version 1.3.2⁵⁴.

Scoring

During the main search, many attributes of the fitting process are collected for each precursor. These include the retention time error, the fraction of the library intensity that is matched, and the scribe score,⁵⁵ which are later incorporated into the scoring model.

Every coefficient and its corresponding attributes from the main search are collected and saved. In this output, each row is a unique precursor in a specific spectrum. These data are then grouped

by precursor to give a list of potential precursor identifications, both targets and decoys. By default, random forest models, implemented using scikit-learn⁵⁶, are trained with 5-fold cross-validation to differentiate between targets and decoys. Predictions for each peptide are made with a model that has not seen that peptide before in any charge state. JMod allows users to specify the preferred model and choose from 4 different built-in models or add their own. It also provides all data used to train the models so that users can reanalyze their data if they wish to add additional features or external scoring models.

Quantitation

By default, JMod reports the maximum coefficient predicted over all possible scans for MS2 quantitation (*coeff*). This is analogous to quantitation with a scan window range of 1. Users can also extract more intricate measures from the output files, as these include all predicted coefficients for each precursor in each spectrum. The MS2 quantities reported in the paper use the maximum coefficient.

JMod provides more options by default for MS1 quantitation. For label-free searches, MS1 quantities are reported as both the area underneath the monoisotopic peak surrounding the best matching coefficient (*MS1 Area*) as well as the coefficient when fitting the theoretical isotopic pattern to the MS1 scan with the maximum monoisotopic intensity (*MS1 Int*). When using non-isobaric mass tags and plexDIA, JMod makes use of the inherent structure provided by the different channels. All channels within a precursor set are fit together. First, the most confident coefficient is selected for each channel. These are allowed to occur in different scans, as multiplexing using deuterium induces a retention time shift³⁷. The resulting retention time range defines a set of MS1 scans over which all channels and their theoretical isotopic envelopes are fit to the observed signal, giving a series of MS1 coefficients that define the MS1 elution peak of each channel. JMod returns the area under these peaks as *plex Area* as well as the maximum value as *plexfitMS1*. We found *plex Area* provided the best accuracy, while *plexfitMS1* reported better precision. If one is using tags that do not induce retention time shifting, one can enforce quantitation from the same scan(s) for better results.

When using non-isobaric mass tags, each channel is scored independently. However, each pre-

cursor is assigned a value *BestChannel_Qvalue* which is the q-value of the best scoring channel in that set. Protein q-values are calculated by assigning the maximum target or decoy score to each respective protein. The proteins are then ranked according to this score. Protein q-values are estimated using the fraction of proteins with a higher score which are decoys.

Entrapment

The entrapment FDRs were calculated using the paired method from Wen et al³⁶. Briefly, for each target sequence in the database, we procured a shuffled entrapment sequence by randomly permuting all but the last amino acid in the target sequence. If after 20 shuffling attempts no entrapment sequence was found that was unique from the original target sequences, both that target and the corresponding entrapment sequence were removed from the database. There was therefore a one-to-one correspondence between the original targets and their shuffled entrapment sequences. Decoy sequences were generated from the original and entrapment targets using the reversal method. The library fragment *m/z* ratios were adjusted according to the permutation, that is shuffling or reversal, and the intensities remained the same as the parent target sequences. For the paired entrapment method, Wen et al. proposed the following formula:³⁶

$$FDP = \frac{N_e + N_{e \geq s > t} + 2 \cdot N_{e > t \geq s}}{N_t + N_e}$$

where N_e and N_t , respectively, represent the number of entrapment targets and original targets scoring better than the current score threshold, s . $N_{e \geq s > t}$ indicates the number of entrapment targets at s for which their scores are greater than or equal to s but the score of their respective original complements are lower than s . Likewise $N_{e > t \geq s}$ indicates the number of entrapment targets for which their scores were greater than their respective original targets, and for which their original targets also scored greater than or equal to s .

DIA-NN parameters

DIA-NN version 1.9 was used in the label-free entrapment analysis. This version provides internal scores (CScores) that can be used to calculate the entrapment FDR. For these analyses the data

were searched with match between runs turned off, run-to-run normalization disabled, protein inference turned off, the precursor FDR(%) set to 50, and with otherwise default parameters.

DIA-NN version 2.1 was used to search the PSM-tag data. PSMtag was added as a fixed modification using `{-fixed-mod tag, 308.1160923903, nK -channels tag,d0,nK,0:0; tag,d4,nK,4.01096:4.01096; tag,d8,nK,8.026839:8.026839; tag,d12,nK,12.033938:12.033938; tag,d16,nK,16.038574:16.038574 -original-mods }`. Quantitation was set to `-high-acc`.

For diethylation, the same parameters were used with channels defined using the following commands; `{-fixed-mod Diethyl, 56.06260026, nK -lib-fixed-mod Diethyl -channels Diethyl,0,nK,0:0; Diethyl,4,nK,4.01342:4.01342; Diethyl,8,nK,8.05021:8.05021; -original-mods }`

Data filtering

PSMtag identification rates for DIA-NN are shown for $Q.Value < 0.01$ with no $Channel.Q.Value$ filtering. JMod identifications for the same data are shown with the equivalent $BestChannel.Qvalue$ of < 0.01 . Protein level data for both platforms are filtered with a library protein q-value of 0.01. These were calculated using the best protein q-value across all replicates for each algorithm.

For quantitative comparisons, MS1 quantities for DIA-NN are shown using the $Ms1.Area$ of precursors with a $Q.Value < 0.01$ and $Channel.Q.Value < 0.01$. For JMod, MS1 quantitation is shown using $plex.Area$ and $Qvalue < 0.01$.

MS2 quantitation for JMod is shown using the maximum coefficient ($coeff$) and only for precursors with at least 3 b-ions. Arginine-terminating precursors from different channels without any matched b-ions will have the same template in the fitting process and thereby result in equal coefficients.

References

1. Cravatt, B. F., Simon, G. M. & Yates Iii, J. R. The biological impact of mass-spectrometry-based proteomics. *Nature* **450**, 991–1000 (2007).
2. MacCoss, M. J. *et al.* Sampling the proteome by emerging single-molecule and mass spectrometry methods. en. *Nat. Methods* **20**, 339–346. <https://www.nature.com/articles/s41592-023-01802-5> (Mar. 2023).
3. Matthews, M. L. *et al.* Chemoproteomic profiling and discovery of protein electrophiles in human cells. *Nature chemistry* **9**, 234–243 (2017).
4. Budnik, B., Levy, E., Harmange, G. & Slavov, N. SCoPE-MS: mass-spectrometry of single mammalian cells quantifies proteome heterogeneity during cell differentiation. *Genome Biology* **19**, 161. <https://doi.org/10.1186/s13059-018-1547-5> (2018).
5. Johnston, S. M. *et al.* Rapid, One-Step Sample Processing for Label-Free Single-Cell Proteomics. en. *J. Am. Soc. Mass Spectrom.* ISSN: 1044-0305, 1879-1123. <https://pubs.acs.org/doi/10.1021/jasms.3c00159> (July 2023).
6. Gatto, L. *et al.* Initial recommendations for performing, benchmarking, and reporting single-cell proteomics experiments. *Nat. Methods* **20**, 375–386. <https://doi.org/10.1038/s41592-023-01785-3> (2023).
7. Derks, J. *et al.* Single-nucleus proteomics identifies regulators of protein transport. *bioRxiv*. <https://doi.org/10.1101/2024.06.17.599449> (2024).
8. Leduc, A., Khoury, L., Cantlon, J., Khan, S. & Slavov, N. Massively parallel sample preparation for multiplexed single-cell proteomics using nPOP. *Nature Protocols*, 1–27. <https://doi.org/10.1038/s41596-024-01033-8> (2024).
9. Leduc, A., Xu, Y., Shipkovenska, G., Dou, Z. & Slavov, N. Limiting the impact of protein leakage in single-cell proteomics. *Nature Communications* **16**, 4169 (2025).
10. Messner, C. B. *et al.* Mass spectrometry-based high-throughput proteomics and its role in biomedical studies and systems biology. *Proteomics* **23**, 2200013 (2023).
11. Slavov, N. Increasing proteomics throughput. *Nature Biotechnology* **39**, 809–810. <https://doi.org/10.1038/s41587-021-00881-z> (2021).
12. Venable, J. D., Dong, M.-Q., Wohlschlegel, J., Dillin, A. & Yates, J. R. Automated approach for quantitative analysis of complex peptide mixtures from tandem mass spectra. en. *Nat. Methods* **1**, 39–45 (Oct. 2004).
13. Dong, M.-Q. *et al.* Quantitative Mass Spectrometry Identifies Insulin Signaling Targets in *C. elegans*. *Science* **317**, 660–663 (2007).
14. Doerr, A. DIA mass spectrometry. en. *Nat. Methods* **12**, 35–35 (Jan. 2015).
15. Ludwig, C. *et al.* Data-independent acquisition-based SWATH-MS for quantitative proteomics: a tutorial. *Molecular systems biology* **14**, e8126 (2018).
16. Tsou, C.-C. *et al.* DIA-Umpire: comprehensive computational framework for data-independent acquisition proteomics. *Nature methods* **12**, 258–264 (2015).

17. Sinitcyn, P. *et al.* MaxDIA enables library-based and library-free data-independent acquisition proteomics. *Nature Biotechnology*, 1–11 (2021).
18. Wallmann, G. *et al.* AlphaDIA enables end-to-end transfer learning for feature-free proteomics. *bioRxiv*, 2024–05 (2024).
19. Demichev, V. *et al.* dia-PASEF data analysis using FragPipe and DIA-NN for deep proteomics of low sample amounts. *Nature communications* **13**, 3944 (2022).
20. Szyrwił, L., Sinn, L., Ralser, M. & Demichev, V. Slice-PASEF: fragmenting all ions for maximum sensitivity in proteomics. *BioRxiv*, 2022–10 (2022).
21. Demichev, V., Messner, C. B., Vernardis, S. I., Lilley, K. S. & Ralser, M. DIA-NN: neural networks and interference correction enable deep proteome coverage in high throughput. *Nature methods* **17**, 41–44 (2020).
22. Searle, B. C. *et al.* Chromatogram libraries improve peptide detection and quantification by data independent acquisition mass spectrometry. en. *Nat. Commun.* **9**, 5128 (Dec. 2018).
23. Bruderer, R. *et al.* Extending the limits of quantitative proteome profiling with data-independent acquisition and application to acetaminophen-treated three-dimensional liver microtissues. *Molecular & Cellular Proteomics* **14**, 1400–1410 (2015).
24. Peckner, R. *et al.* Specter: linear deconvolution for targeted analysis of data-independent acquisition mass spectrometry proteomics. *Nature methods* **15**, 371–378 (2018).
25. Frejno, M. *et al.* Unifying the analysis of bottom-up proteomics data with CHIMERYs. *bioRxiv*, 2024.05.27.596040 (May 2024).
26. Derks, J. *et al.* Increasing the throughput of sensitive proteomics by plexDIA. *Nature Biotechnology*. <https://doi.org/10.1038/s41587-022-01389-w> (2022).
27. Derks, J., McDonnell, K., Wamsley, N., Specht, H. & Slavov, N. Increasing mass spectrometry throughput using time-encoded sample multiplexing. *bioRxiv* (2025).
28. Thielert, M. *et al.* Robust dimethyl-based multiplex-DIA doubles single-cell proteome depth via a reference channel. *Molecular Systems Biology* **19**, e11503 (2023).
29. Framework for multiplicative scaling of single-cell proteomics. en. *Nat. Biotechnol.*, 1–2. <https://www.nature.com/articles/s41587-022-01411-1> (July 2022).
30. Derks, J. & Slavov, N. Strategies for increasing the depth and throughput of protein analysis by plexDIA. *Journal of Proteome Research* **22**, 697–705. <https://doi.org/10.1021/acs.jproteome.2c00721> (2023).
31. Specht, H. *et al.* PSMtags improve peptide sequencing and throughput in sensitive proteomics. *bioRxiv* (2025).
32. Choi, Y. *et al.* MS1-level proteome quantification platform allowing maximally increased multiplexity for SILAC and in vitro chemical labeling. *Analytical Chemistry* **92**, 4980–4989 (2020).
33. Peterson, A. C., Russell, J. D., Bailey, D. J., Westphall, M. S. & Coon, J. J. Parallel reaction monitoring for high resolution and high mass accuracy quantitative, targeted proteomics. en. *Mol. Cell. Proteomics* **11**, 1475–1488 (Nov. 2012).

34. Guzman, U. H. *et al.* Ultra-fast label-free quantification and comprehensive proteome coverage with narrow-window data-independent acquisition. *Nature biotechnology* **42**, 1855–1866 (2024).
35. Slavov, N. Driving Single Cell Proteomics Forward with Innovation. *Journal of Proteome Research* **20**, 4915–4918. <https://doi.org/10.1021/acs.jproteome.1c00639> (2021).
36. Wen, B. *et al.* Assessment of false discovery rate control in tandem mass spectrometry analysis using entrapment. *bioRxiv* (2024).
37. Boutilier, J. M., Warden, H., Doucette, A. A. & Wentzell, P. D. Chromatographic behaviour of peptides following dimethylation with H₂/D₂-formaldehyde: implications for comparative proteomics. en. *J. Chromatogr. B Analyt. Technol. Biomed. Life Sci.* **908**, 59–66 (Nov. 2012).
38. Burnum-Johnson, K. E. *et al.* New views of old proteins: clarifying the enigmatic proteome. *Molecular & Cellular Proteomics* **21**, 100254 (2022).
39. Liang, Y. *et al.* HyperSCP: combining isotopic and isobaric labeling for higher throughput single-cell proteomics. *Analytical chemistry* **95**, 8020–8027 (2023).
40. Slavov, N. Scaling Up Single-Cell Proteomics. *Molecular & Cellular Proteomics* **21**, 100179. ISSN: 1535-9476. <https://doi.org/10.1016/j.mcpro.2021.100179> (2022).
41. Tsour, S., Machne, R., Leduc, A., Widmer, S. & Slavov, N. Alternate RNA decoding results in stable and abundant proteins in mammals. *bioRxiv*. <https://doi.org/10.1101/2024.08.26.609665> (2024).
42. Bortçen, T., Müller, T. & Krijgsveld, J. An integrated workflow for quantitative analysis of the newly synthesized proteome. *Nature Communications* **14**, 8237 (2023).
43. Leduc, A. & Slavov, N. Protein degradation and growth dependent dilution substantially shape mammalian proteomes. *bioRxiv*, 2025–02. <https://doi.org/10.1101/2025.02.10.637566> (2025).
44. Gessulat, S. *et al.* Prosit: proteome-wide prediction of peptide tandem mass spectra by deep learning. *Nature methods* **16**, 509–518 (2019).
45. Tiwary, S. *et al.* High-quality MS/MS spectrum prediction for data-dependent and data-independent acquisition data analysis. *Nature methods* **16**, 519–525 (2019).
46. Picciani, M. *et al.* Oktoberfest: Open-source spectral library generation and rescoring pipeline based on Prosit. *Proteomics*, e2300112 (Sept. 2023).
47. Derks, J. & Slavov, N. Strategies for increasing the depth and throughput of protein analysis by plexDIA. *bioRxiv* 2022.11.05.515287. <https://doi.org/10.1101/2022.11.05.515287> (2022).
48. Jung, J. *et al.* Deuterium-free, three-plexed peptide diethylation for highly accurate quantitative proteomics. *Journal of proteome research* **18**, 1078–1087 (2019).
49. Koehler, C. J., Arntzen, M. Ø. & Thiede, B. The impact of carbon-13 and deuterium on relative quantification of proteins using stable isotope diethyl labeling. en. *Rapid Commun. Mass Spectrom.* **29**, 830–836 (May 2015).

50. Zeng, W.-F. *et al.* AlphaPeptDeep: a modular deep learning framework to predict peptide properties for proteomics. *Nature Communications* **13**, 7238 (2022).
51. Dittwald, P. & Valkenborg, D. BRAIN 2.0: time and memory complexity improvements in the algorithm for calculating the isotope distribution. *Journal of the American Society for Mass Spectrometry* **25**, 588–594 (2014).
52. Abdrakhimov, D. A. *et al.* Biosaur: An open-source Python software for liquid chromatography–mass spectrometry peptide feature detection with ion mobility support. *Rapid Communications in Mass Spectrometry*, e9045 (2021).
53. Elias, J. E. & Gygi, S. P. Target-decoy search strategy for increased confidence in large-scale protein identifications by mass spectrometry. *Nature methods* **4**, 207–214 (2007).
54. Andersen, M. S., Dahl, J. & Vandenberghe, L. *CVXOPT: A Python package for convex optimization, version 1.3.2* Available at cvxopt.org. 2023.
55. Searle, B. C., Shannon, A. E. & Wilburn, D. B. Scribe: next generation library searching for DDA experiments. *Journal of Proteome Research* **22**, 482–490 (2023).
56. Pedregosa, F. *et al.* Scikit-learn: Machine learning in Python. *the Journal of machine Learning research* **12**, 2825–2830 (2011).